EECE 2311

Lab for EECE2310: Introduction to Digital Design and Computer Architecture

SPRING 2026

Lab Assignment #03

John Kimani

EECE2311 Lab Section # 33858

Sijing Deng

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Abstract

Designing Binary Coded Decimal (BCD) decoder, Binary Coded Decimal (BCD) decoder with enable and negative outputs, Multiple Binary Coded Decimal (BCD) decoder with controlling 7-segment displays with enable.

Introduction

A seven-segment display is an electronic component that can be used to display a 1-digit number from 0-9 and a letter A to F. In this lab we will design a digital circuit to display digits 0 to 9 only, as shown in Figures 1. The circuit we will design is called a Binary Coded Decimal (BCD) decoder, Figures 2. This circuit takes in a 4-bit input representing the binary equivalent of a digit 0 to 9 and lights up the desired segments to display the corresponding digit. You are not allowed to use any of the Quartus software library modules unless for the modules that are explicitly allowed in the lab instructions.



Figure 1: 7-segment display output digits

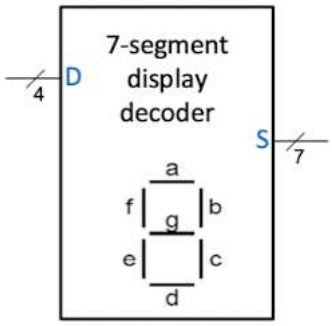


Figure 2: BCD Decoder

Lab Setup

Pre-Lab

The DE1-SoC FPGA circuit turns on any segment in the 7-segment display by applying logic 0 to that segment (i.e., active low) . To turn it off, apply logic 1. This is because the 7-segment LED displays have a common anode, which means the anode (positive) side of all of these LEDs are electrically connected together to the Vcc (logic 1), and each LED cathode (negative) side is connected to the input pin that drives the LED, as shown in Figure 3.

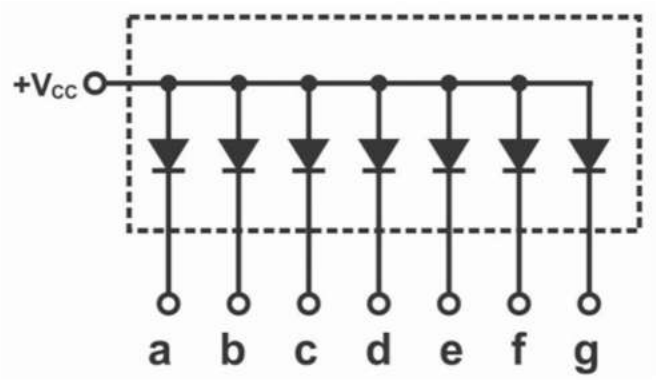


Figure 3: Common anode LEDs configuration

1. Complete the table below to indicate which segments will be turned ON (0) vs OFF (1) for the digits 0 to 9 shown in Figure 1 above. The entries for digit 0 have been done for you.

Decimal #	BCD Inputs				7-Segment Outputs						
	D_3	D_2	D_1	D_0	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	0	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0

Table 1: Digits to display on the 7-segment display

2. Use seven Karnaugh maps to find the minimized Boolean expressions for each of the seven outputs shown in Table 1 above (outputs a to g).

D_3D_2 \ D_1D_0	00	01	11	10
00	0	1	0	0
01	1	0	0	0
11				
10	0	0		

Table 2: Karnaugh map for segments a

$a = D_3'D_2'D_1'D_0 + D_2'D_1'D_0'$

D_3D_2 \ D_1D_0	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11				
10	0	0		

Table 3: Karnaugh map for segments b

$b = D_2'D_1D_0 + D_2D_1D_0'$

D_3D_2 \ D_1D_0	00	01	11	10
00	0	0	0	1
01	0	0	0	0
11				
10	0	0		

Table 4: Karnaugh map for segments c

$c = D_2'D_1D_0'$

D_3D_2 \ D_1D_0	00	01	11	10
00	0	1	0	0
01	1	0	1	0
11				
10	0	0		

Table 5: Karnaugh map for segments d

$d = D_2D_1'D_0' + D_2D_1D_0 + D_3'D_2'D_1'D_0$

D_3D_2 \ D_1D_0	00	01	11	10
00	0	1	1	0
01	1	1	1	0
11				
10	0	1		

Table 6: Karnaugh map for segments e

$e = D_0 + D_2D_1'$

D_3D_2 \ D_1D_0	00	01	11	10
00	0	1	1	1
01	0	0	0	0
11				
10	0	0		

Table 7: Karnaugh map for segments f

$f = D_2'D_1 + D_3'D_2'D_0$

D_3D_2 \ D_1D_0	00	01	11	10
00	1	1	0	0
01	0	0	1	0
11				
10	0	0		

Table 7: Karnaugh map for segments g

$g = D_3'D_2'D_1' + D_2D_1D_0$

Equipment

- DE1-SoC FPGA development board
- Computer with Intel Quartus Prime software installed
- USB programming cable for FPGA configuration
- Previously designed half-adder and full-adder schematic files
- Connecting cables and power supply for the board

2. Testing the BCD Decoder on the DE1-SoC Board

Verify the correct behavior and output of your BCD to 7-segment decoder design on the display by changing the switches setting.

Assignment 1

Verify the correct behavior and output of your BCD to 7-segment decoder design on the display by changing the switches settings. Take pictures of your Board's switches and 7-segment displays with at least six different settings for the switches and show the corresponding output on the display. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

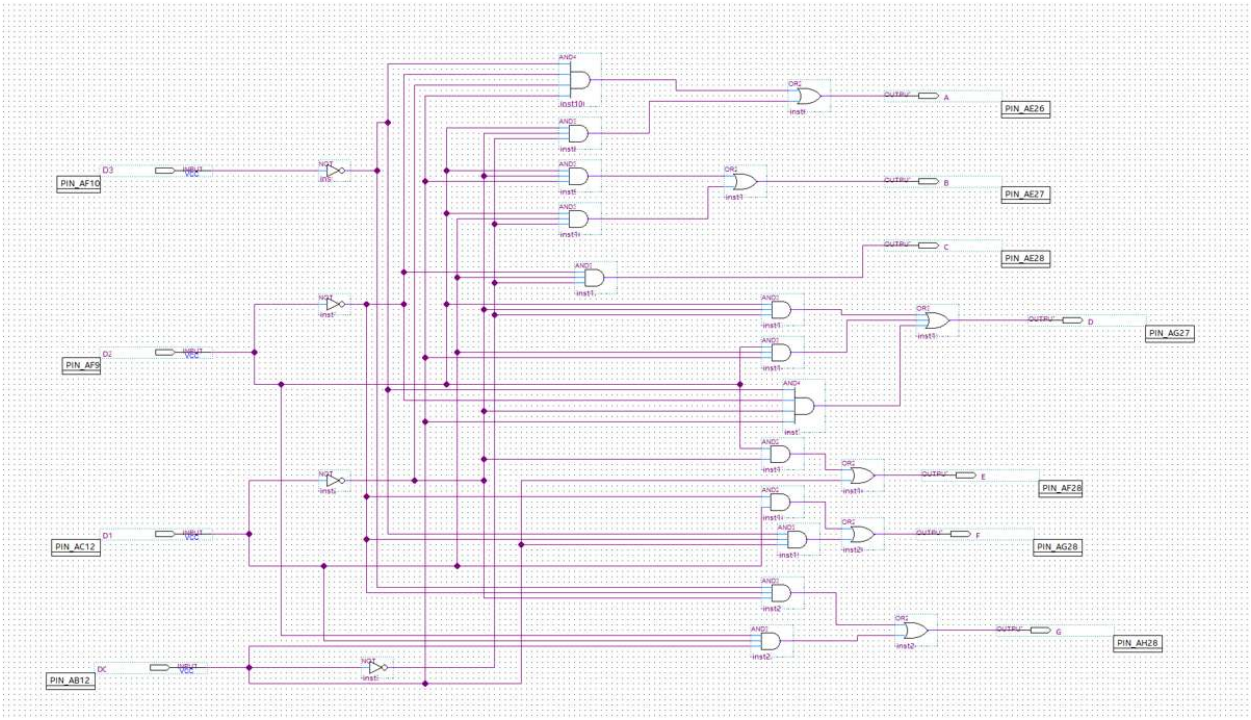


Figure 4: display7 assignment 1 schematic

The BCD decoder translates this 4-bit input into seven output signals, each controlling one segment of the display. Because the DE1-SoC board uses a **common-anode** seven-segment display, the segments are **active low**—meaning a logic 0 turns a segment ON and a logic 1 turns it OFF. Electrically, all LED anodes are tied together to Vcc (logic 1), and each segment's cathode is connected to a decoder output pin. When the decoder outputs 0, current flows from Vcc through the LED segment to the FPGA pin, illuminating that segment.

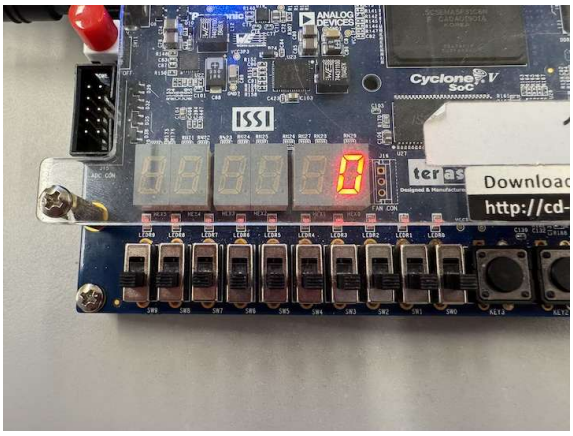


Figure 5: display 7 decimal 0 (0000)

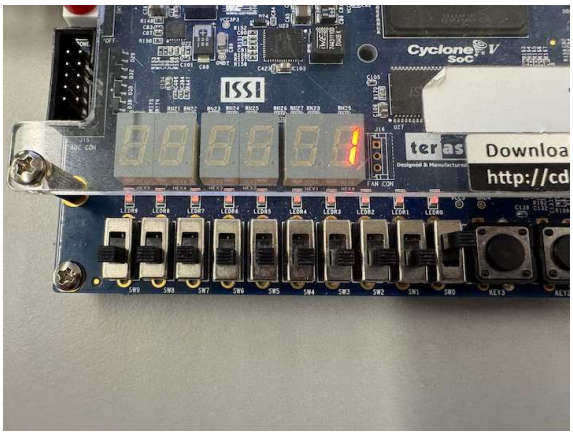


Figure 6: display 7 decimal 1 (0001)

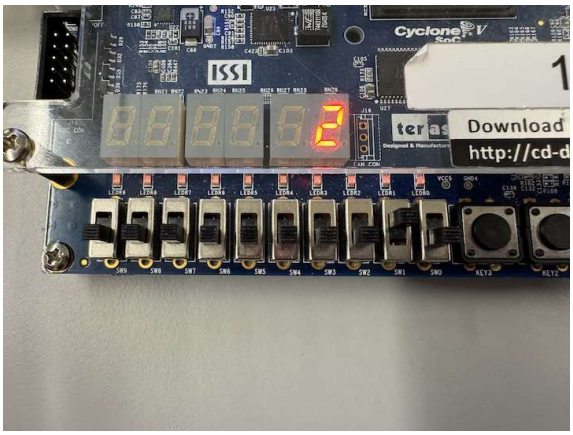


Figure 8: display 7 decimal 2 (0010)

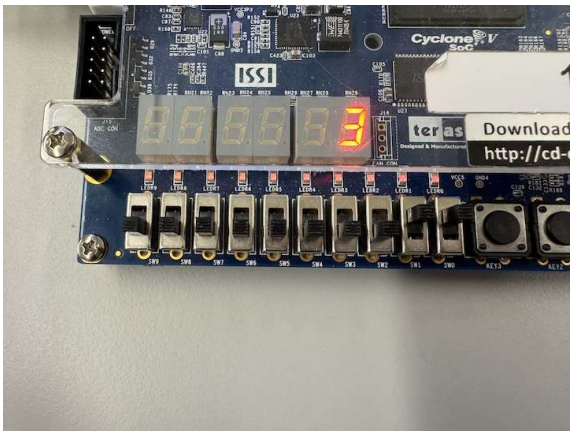


Figure 9: display 7 decimal 3 (0011)

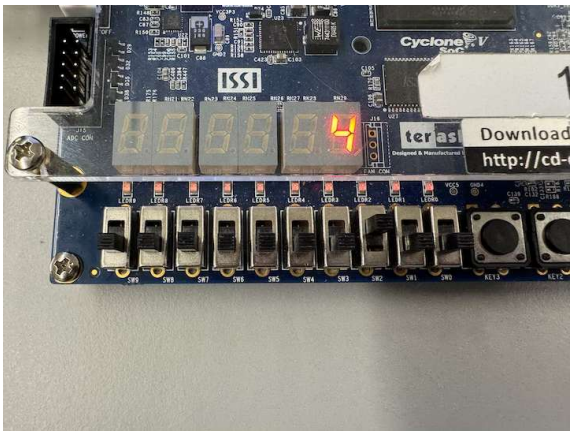


Figure 10: display 7 decimal 4 (0100)

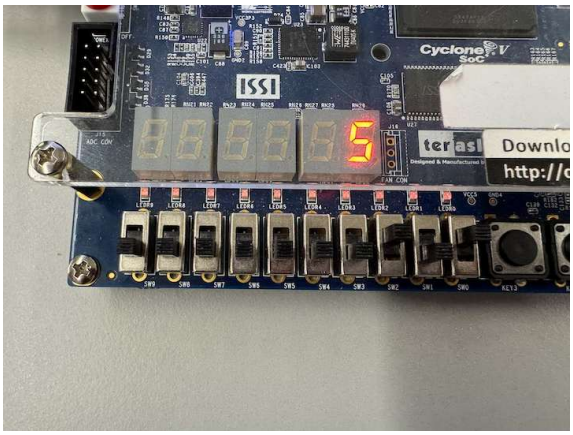


Figure 11: display 7 decimal 5 (0101)

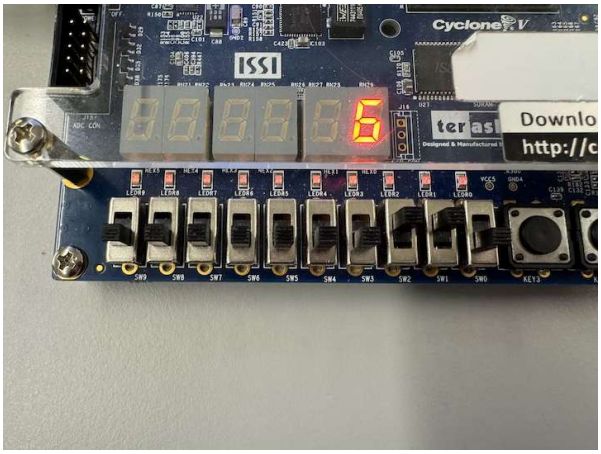


Figure 12: display 7 decimal 6 (0110)

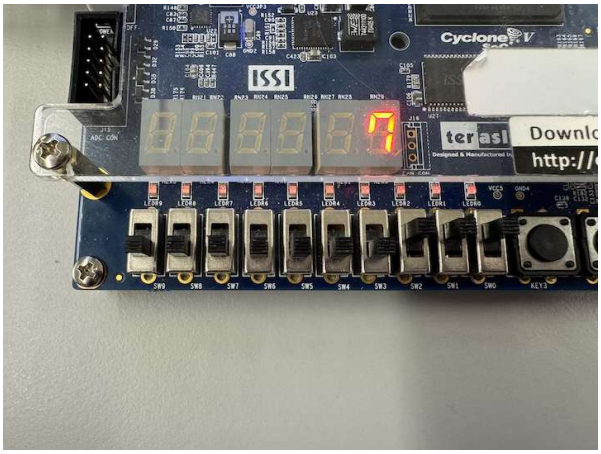


Figure 13: display 7 decimal 7 (0111)

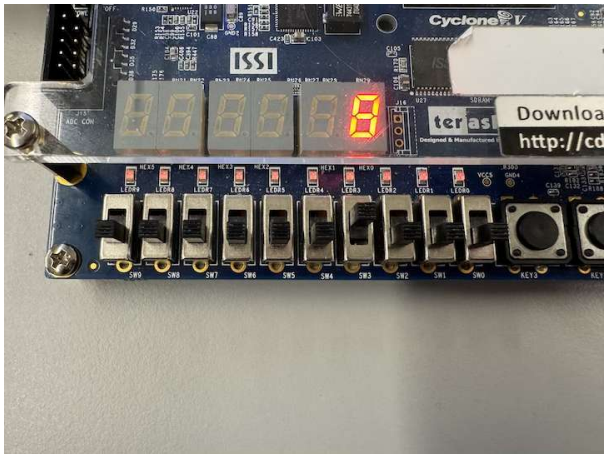


Figure 14: display 7 decimal 8 (1000)

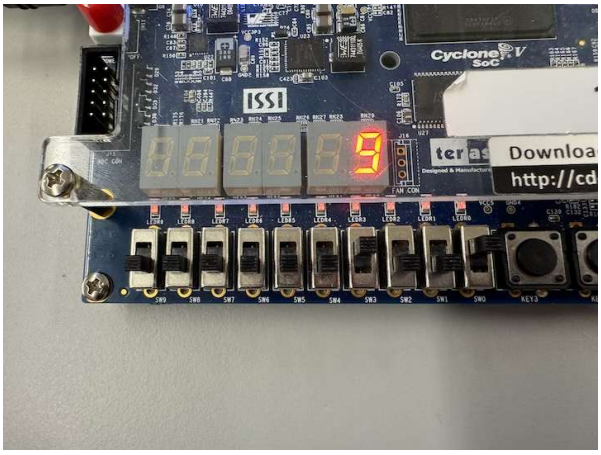


Figure 15: display 7 decimal 9 (1001)

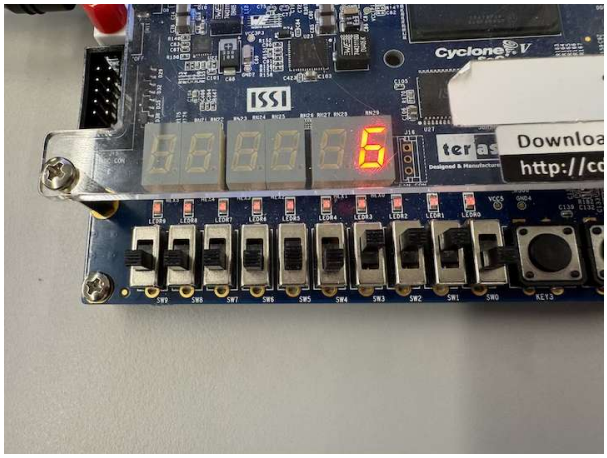


Figure 16: Random decimal 6 (1110)

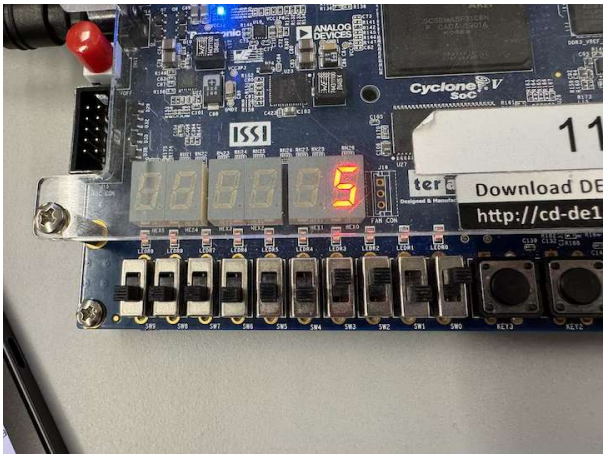


Figure 16: Random decimal 5 (1101)

3. 7-Segment Display with Enable and Negative Inputs

Modify the 7-segment decoder to have two new inputs: En and Ng as shown in Figure 4. In the current implementation, when all the four switches of the D input are OFF, representing number zero, the display shows the digit '0' which is the expected behavior so far. In some applications when the display is not in use, we want it to have all its segments turned OFF. The new En (enable) input to our BCD decoder digital circuits has been added to help us with this option. When the En switch input is ON (i.e., binary 1), the 7-Segment display must work as usual by displaying the decimal digit corresponding to 4-bit binary input D. When the En input is OFF (i.e., binary 0), all displays must be turned OFF and must not show anything regardless of the value on input D.

Assignment 2

Take a screenshot of your modified design and include it in the lab report. Take pictures of your Board's switches (including the enable and negative switches) and the 7-segment displays with at least four different settings of the switches, which must show the effect of the ***En*** and ***Ng*** inputs, and the corresponding outputs on the display. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

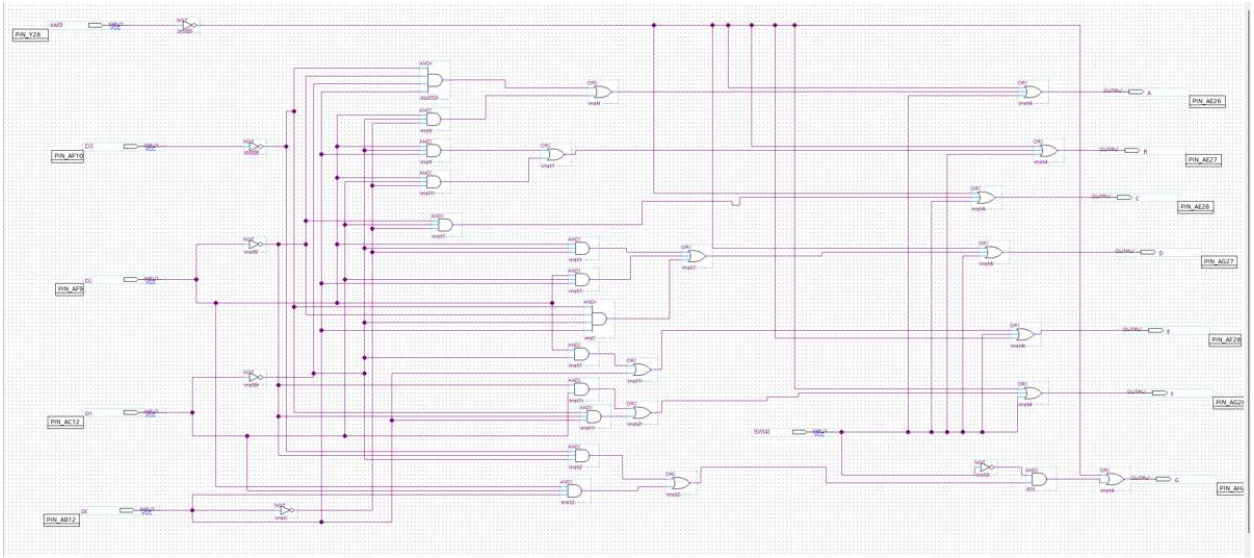


Figure 16: display 7 encoder/ negative schematic

The provided schematic illustrates a modified **BCD to 7-segment decoder** designed for the DE1-SoC FPGA, which translates a 4-bit binary input (D3 to D0) into the specific boolean functions required to drive segments **a** through **g**. The architecture employs a combination of

inversion, product-term generation (AND gates), and summation (OR gates) to map binary values to their decimal visual equivalents.

- **Enable Logic (SW[9]):** This input serves as a master gate for the entire output stage. When the **En** signal is logic '0', the implementation ensures that all output lines (A–G) are driven low, effectively blanking the display.
- **Negative Logic (SW[4]):** This signal is designed to override the numerical input. When **Ng** is asserted (logic '1'), the logic suppresses segments A through F while forcing segment **g** (the center bar) to remain active, thereby representing a negative sign.

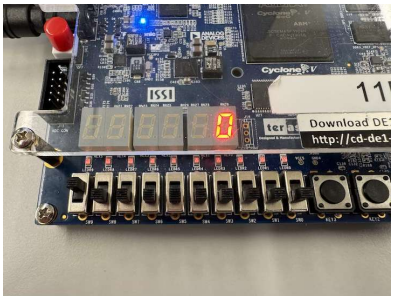


Figure 17: encoder on display 0

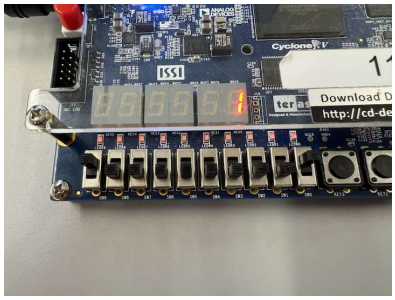


Figure 18: encoder on display 1

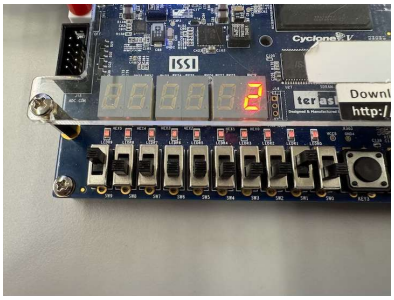


Figure 19: encoder on display 2

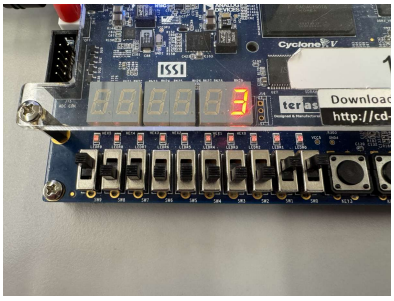


Figure 20: encoder on display 3

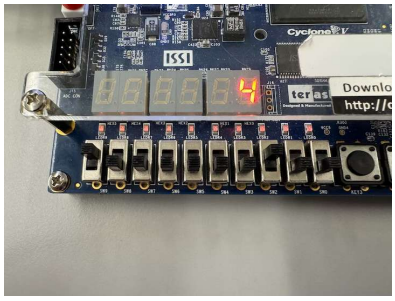


Figure 21: encoder on display 4

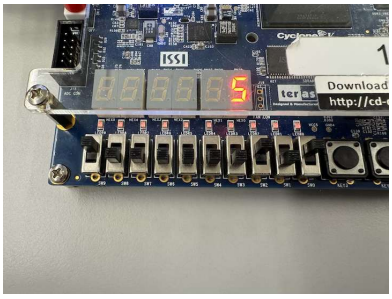


Figure 22: encoder on display 5



Figure 23: encoder on display 6

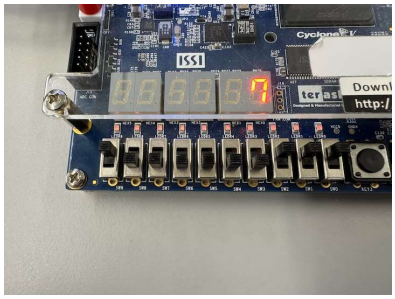


Figure 24: encoder on display 7

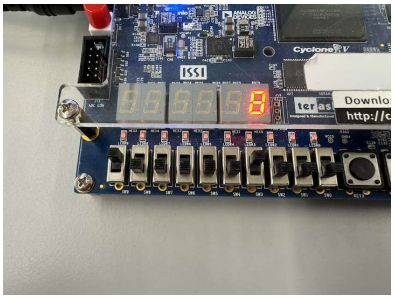


Figure 25: encoder on display 8

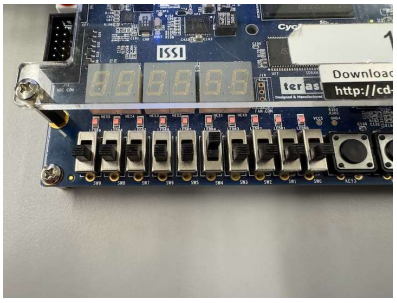


Figure 25: encoder off & negative on display 8



Figure 26: encoder off & negative on display 1

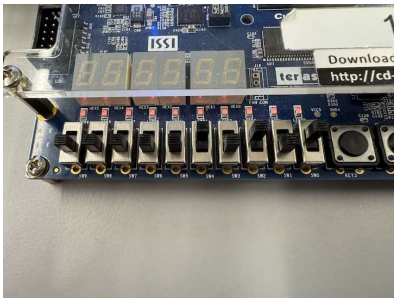


Figure 27: encoder off & negative on display 5

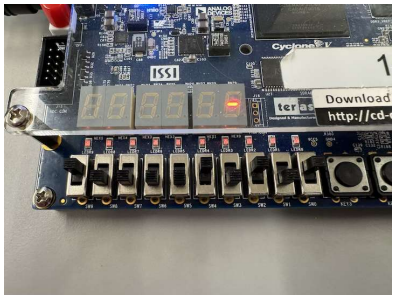


Figure 28: encoder on & negative on display 5

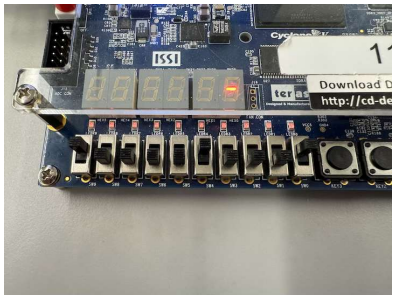


Figure 29: encoder on & negative on display 1



Figure 30: encoder on & negative on display 2



Figure 31: encoder on & negative on display 3



Figure 32: encoder on & negative on display 4

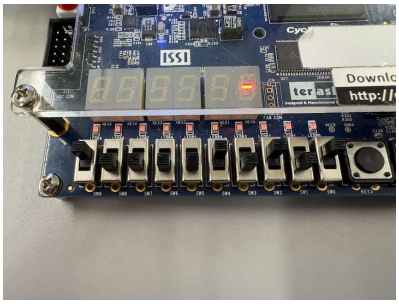


Figure 33: encoder on & negative on display 5

4. Controlling Multiple 7-Segment Displays with Enable

The DE1-SoC features six 7-Segment displays. In this part of the lab, you will display a digit on two 7-segment displays if they are enabled. The digit will be represented by the first four switches SW[3:0], while SW[4] will be used for the negative sign display. Switches SW[8] and SW[9] will be used to enable the 7-segment displays HEX0 and HEX1 respectively.

Assignment 3

Verify the correct behavior of the new design and its outputs on the displays by changing the inputs on the switches for the digits, sign, and enable inputs. Take pictures of your Board's switches and 7-segment displays with at least six different settings for the switches and corresponding output on the displays. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

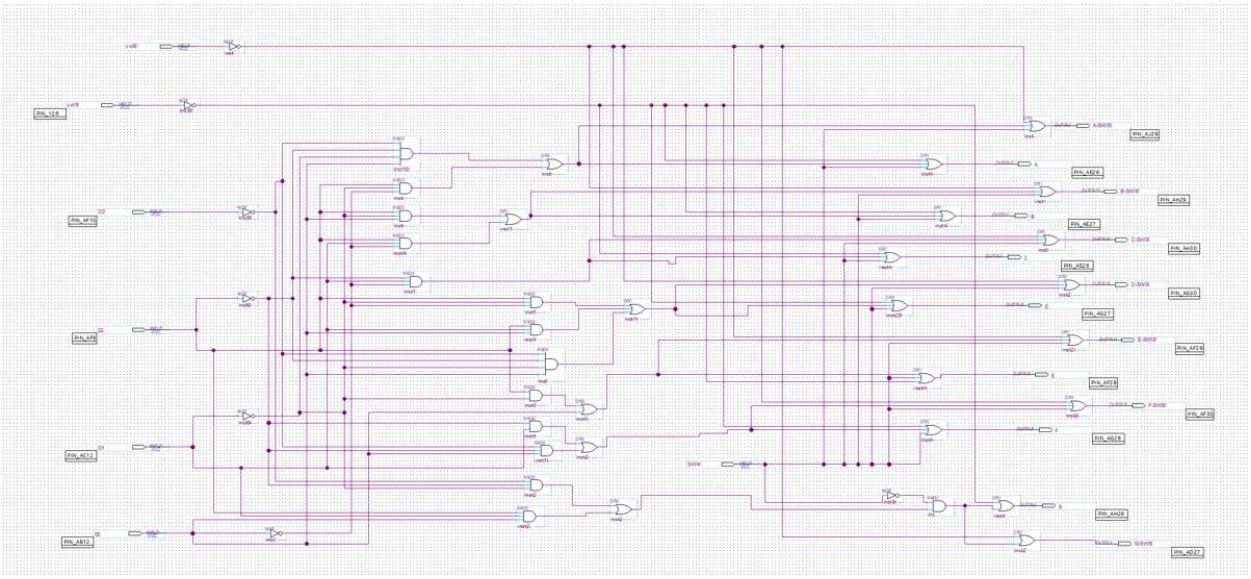


Figure 34: encoder & negative Multiple 7-Segment Displays with Enable

The schematic in **Figure 34** illustrates a combinational logic circuit designed to control multiple **7-segment displays** on the DE1-SoC board, specifically integrating a BCD-to-7-segment decoder with conditional enable and sign-handling logic. The design utilizes input switches **SW[3:0]** to define the numerical digit, which is then processed through a network of logic gates to drive the individual segments (A through G) of the displays. To manage multiple outputs efficiently, the circuit incorporates **Enable** functionality controlled by **SW[8]** and **SW[9]**; these switches act as gatekeepers, determining whether the decoded digit is projected onto **HEX0**, **HEX1**, or both. **SW[4]** allows for the representation of negative signs, effectively expanding the display's utility for signed integer representation.

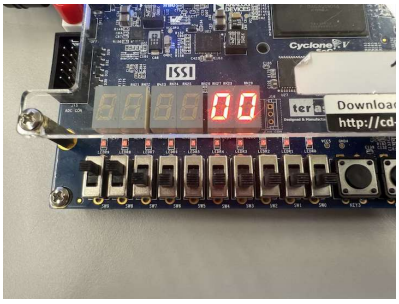


Figure 35: encoder on & negative off display 00

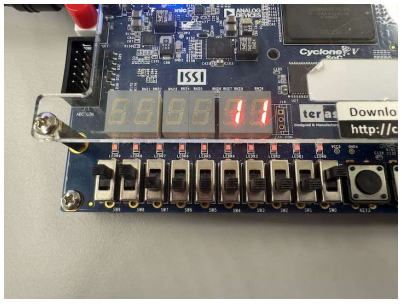


Figure 36: encoder on & negative off display 11

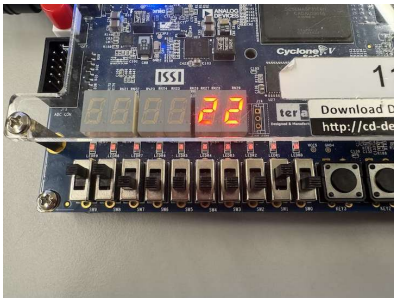


Figure 37: encoder on & negative off display 22

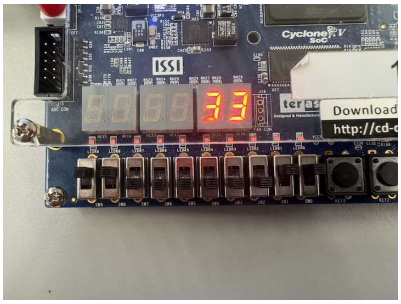


Figure 38: encoder on & negative off display 33

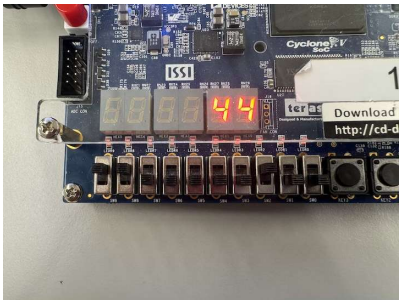


Figure 38: encoder on & negative off display 44

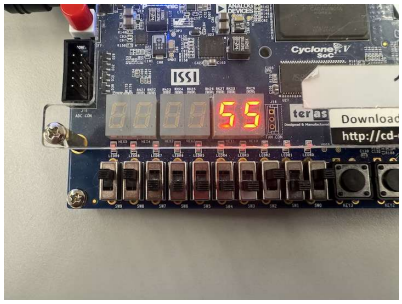


Figure 39: encoder on & negative off display 55

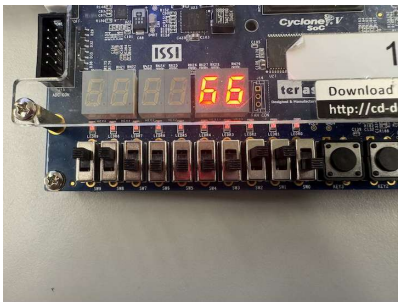


Figure 40: encoder on & negative off display 66

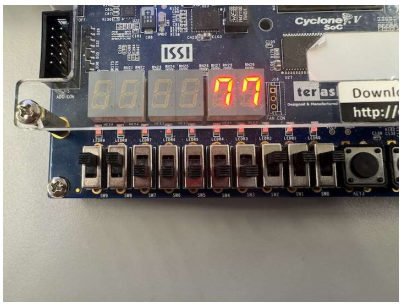


Figure 41: encoder on & negative off display 77

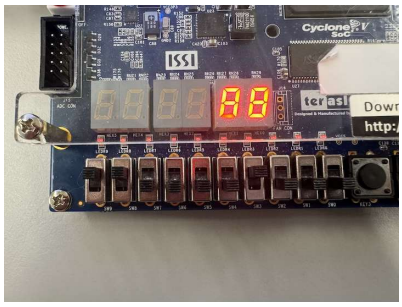


Figure 42: encoder on & negative off display 88

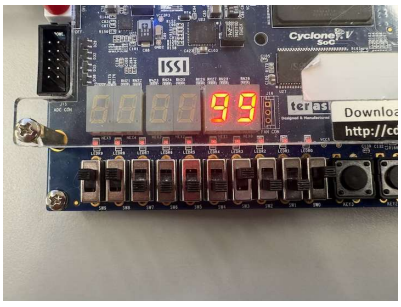


Figure 43: encoder on & negative off display 99



Figure 44: sw[8] on & negative off & sw[9] off display 1

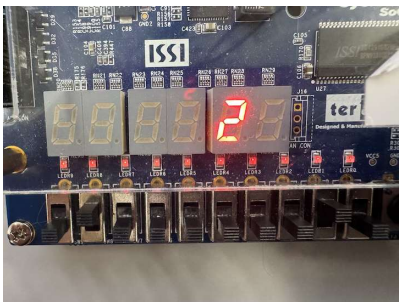


Figure 45: sw[8] on & negative off & sw[9] off display 2

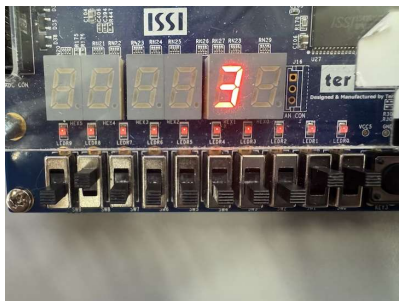


Figure 46: sw[8] on & negative off & sw[9] off display 3

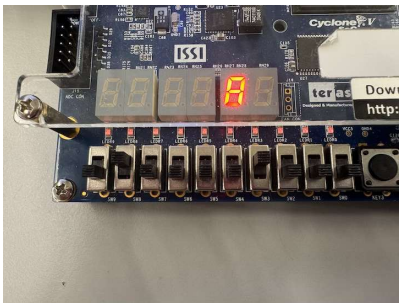


Figure 47: sw[8] on & negative off & sw[9] off display 8

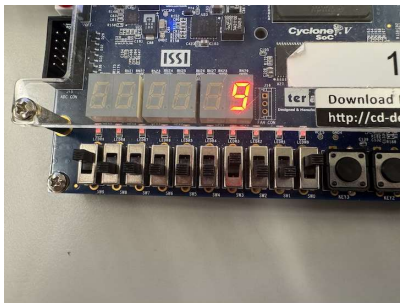


Figure 48: sw[8] off & negative off & sw[9] on display 9

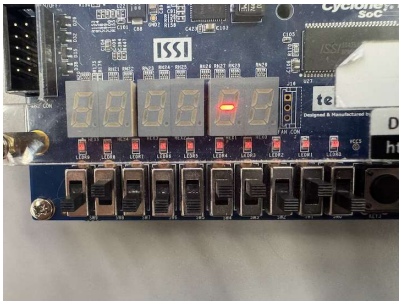


Figure 49: sw[8] on & negative on & sw[9] off display 5

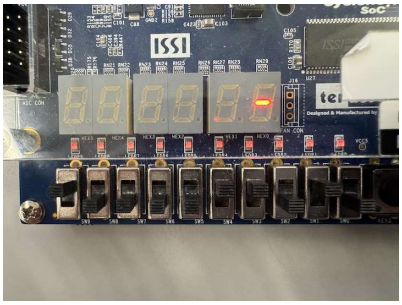


Figure 50: sw[8] off & negative on & sw[9] on display 5

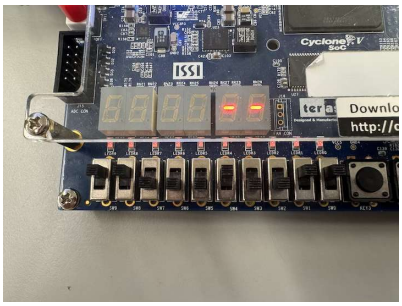


Figure 51: sw[8] on & negative on & sw[9] on display 3

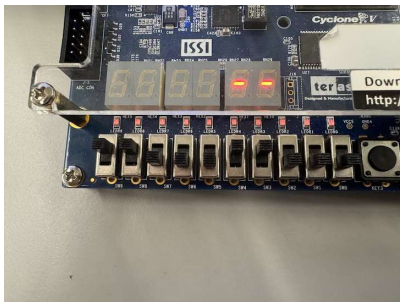


Figure 52: sw[8] on & negative on & sw[9] on display 8

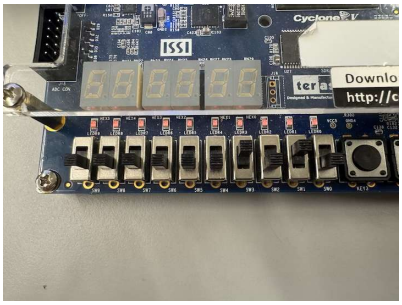


Figure 53: sw[8] off & negative off & sw[9] off display 5

Results and Analysis

Results

The primary objective of this lab was to design and implement a Binary Coded Decimal (BCD) to 7-segment decoder. The results are categorized into three stages of implementation:

1. **Truth Table and K-Maps:** Table 1 was successfully completed, mapping 4-bit binary inputs (0-9) to the active-low outputs (a through g).

2. **Single Digit Decoding:** The decoder was programmed onto the DE1-SoC FPGA. Using switches `SW[3:0]`, the 7-segment display correctly rendered digits 0-9.

3. **Enable and Negative Logic:** The circuit was modified to include an Enable (`En`) and Negative (`Ng`) input.

    - When `En` (SW[9]) was 0, the display remained blank regardless of other inputs.

    - When `Ng` (SW[4]) was 1, the display showed only the center bar (segment g), representing a negative sign.

4. **Multi-Display Control:** The final design successfully controlled `HEX0` and `HEX1` independently using `SW[8]` and `SW[9]` as enable signals, allowing the same digit or sign to be toggled across multiple displays.

Analysis

The implementation behaved exactly as predicted by the theoretical Boolean expressions. Because the DE1-SoC uses a common anode configuration, the logic had to be inverted (active low). A logic '0' completes the circuit to ground, allowing current to flow and light the LED, while a logic '1' creates no potential difference, turning the segment off.

The addition of the Enable and Negative logic demonstrated how combinational logic can be used to override data inputs. By placing an `AND` gate (or equivalent logic) at the output stage, the `En` signal effectively "masked" the BCD data. The Negative logic similarly prioritized the `g` segment while suppressing a-f, showing that hardware level UI constraints (like displaying a minus sign) are easily managed through gate-level design.

Conclusion

This lab successfully demonstrated the transition from Boolean algebra and Karnaugh mapping to physical hardware implementation. We confirmed that a 4-bit binary input can be decoded into a human-readable decimal format using a coordinated network of logic gates.

The project showed the importance of understanding hardware specifications, such as the active-low nature of the DE1-SoC 7-segment displays. Furthermore, the integration of Enable and

Negative controls showed how digital systems manage power (blanking displays) and sign representation.

References

1. Michael Benjamin, "*Lab Report Guide*", Northeastern University, January 18, 2006.
2. EECE 2311 lab03 pdf.